

Randomized and low-rank approximation

Sharp sparsity threshold for injectivity of a random sparse rectangular matrix

Difficulty: challenging **Importance:** interesting to the community

Status: Open

Rating rationale: Challenging because a sharp lower-singular-value threshold needs both necessity and sufficiency; community impact is the sparsity cost of reliable randomized embeddings.

Problem statement

Let $k \rightarrow \infty$ through multiples of 100, $n_k \geq k^{1+c}$ for fixed $c > 0$, and $1 \leq s_k \leq k$. Independently for each column of $M_k \in \mathbb{R}^{k \times n_k}$, choose s_k distinct row locations uniformly and put independent signs $\pm 1/\sqrt{s_k}$ there. Independently select a uniform $k/100$ -element column set I_k .

Determine necessary and sufficient asymptotic conditions on s_k for the existence of an absolute $a > 0$ such that

$$\Pr\{\sigma_{\min}((M_k)_{:I_k}) \geq a\} = 1 - o(1).$$

Probability includes both random constructions. The requested threshold concerns a lower singular-value bound, not an upper distortion estimate.

Reference

Huang, Rudelson, and Tikhomirov, *Well-Invertible Column Subsets of Sparse Matrices Are Rare*, §7, Problem 7.2.

Status check

Searches included "Optimal OSI sparsity" "2026" and the paper title with *threshold*. No sharp threshold result was located. A sufficient condition from a two-sided embedding theorem would not on its own establish necessity here.

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Rechecked [Problem 7.2](#) and [Tikhomirov's later paper](#), which uses different aspect-ratio and randomness regimes. Threshold and sparse-rectangular-matrix searches found no sharp criterion for this fixed-column-sparsity model. A sufficient embedding result alone does not determine the target.